AWS Generative AI Developer – Professional

Course: Software Engineering II

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Weekly Learning Report

- **Focus:** AWS Generative AI Fundamentals and Bedrock; Managing Data for Generative AI
- **Resources:** AWS SkillBuilder Cloud Practitioner Essentials, Udemy AWS Generative AI course
- **Summary of Learning:**

    1. Amazon Bedrock provides fully managed, serverless API access to multiple foundation models (Anthropic, Meta, Amazon Titan, and others) through a single interface — the Converse API — so applications aren't locked into one model provider.
    2. Model customization has two main paths: fine-tuning foundation models on proprietary data for deep behavior change, and LoRA (Low-Rank Adaptation), which trains lightweight adapter layers to achieve similar customization at a fraction of the cost and time.
    3. Retrieval-Augmented Generation (RAG) grounds model responses in an organization's own data by chunking documents, generating vector embeddings, and retrieving relevant context at query time — ideal for frequently changing or proprietary knowledge.
    4. Bedrock Knowledge Bases automate the RAG pipeline end-to-end, including chunking strategy selection, vector store optimization, and built-in tools to evaluate RAG performance.
    5. Multimodal pipelines in Bedrock extend generative AI beyond text, enabling models to process and reason across images and other data types within a single workflow.
    6. Governance and safety are handled through Bedrock Guardrails (content filtering), Automated Reasoning Checks (factual verification), and token-level redaction (masking PII before it reaches the model) critical for enterprise and customer-facing deployments.
    7. Prompt Management and Prompt Flows provide version control and visual orchestration for prompt engineering fundamentals — prompt anatomy, best practices, prompt types, and bias mitigation — directly improve output reliability and reduce hallucination.

8. Enterprise integration and the AWS Well-Architected Generative AI Lens tie these capabilities into a governed, scalable architecture, ensuring generative AI systems are secure, cost-aware, and production-ready.
9. Bedrock Data Automation extracts info from documents/audio/video automatically; SageMaker Data Wrangler cleans and joins tabular data.
10. AWS Glue and Glue Studio run serverless pipelines; Glue Data Quality and CloudWatch metrics catch bad data and job failures early.
11. Amazon Transcribe converts speech to text; Comprehend (and Comprehend Medical) extracts entities, sentiment, and PHI-safe insights from text.
12. Amazon OpenSearch Service is a mature, tunable engine for full-text and vector/RAG search.
13. Amazon S3 Vectors stores and queries embeddings natively in S3 — cheaper than a dedicated vector DB at large scale.
14. RDS pairs with S3 document repositories for grounding; Aurora with the pgvector extension combines transactional and vector search in one database.
15. Effective data management techniques with Amazon S3, including storage classes, lifecycle rules, and cross-region replication for security and cost optimization.

Next week's plan: Agentic AI and Operational Efficiency and Optimization